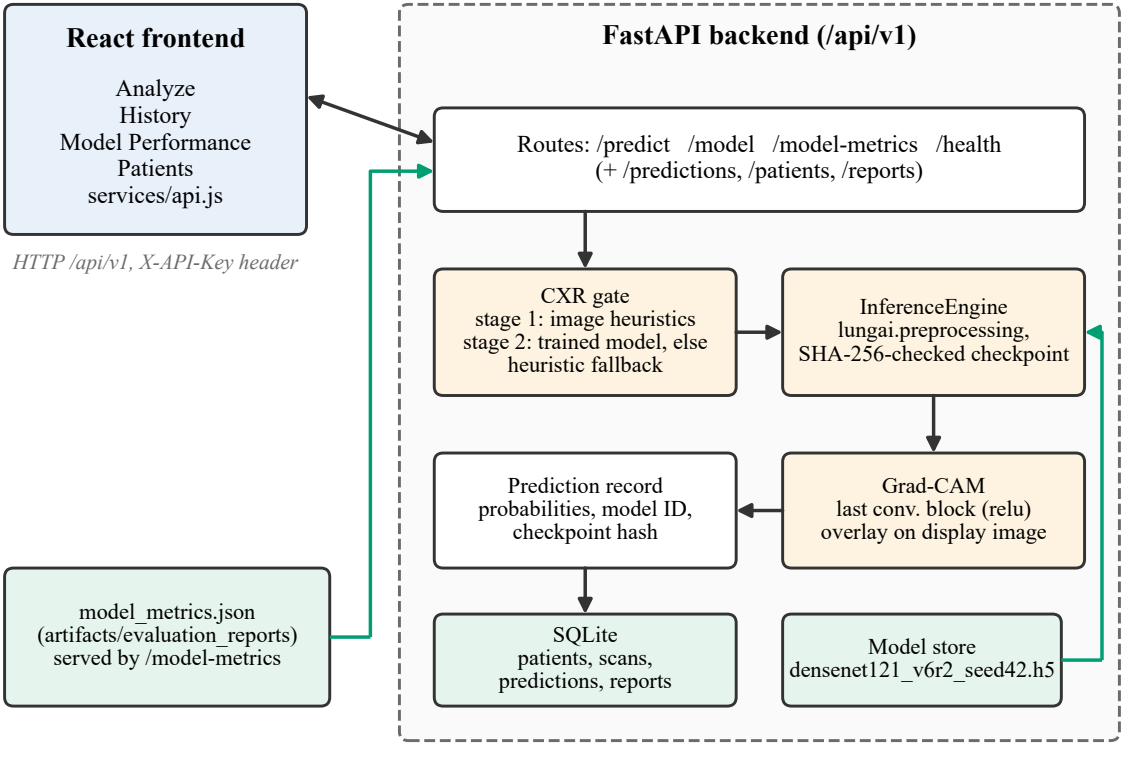

React frontend
Analyze
History
Model Performance
Patients
services/api.js
HTTP /api/v1, X-API-Key header
FastAPI backend (/api/v1)
Routes: /predict /model /model-metrics /health
(+ /predictions, /patients, /reports)
CXR gate
stage 1: image heuristics
stage 2: trained model, else
heuristic fallback
InferenceEngine
lungai.preprocessing,
SHA-256-checked checkpoint
Prediction record
probabilities, model ID,
checkpoint hash
Grad-CAM
last conv. block (relu)
overlay on display image
SQLite
patients, scans,
predictions, reports
Model store
densenet121_v6r2_seed42.h5
model_metrics.json
(artifacts/evaluation_reports)
served by /model-metrics